

Week 4 MG.md

I have fully processed the WEEK-4.pdf slides, merged the core assignment questions and their spin-offs from both the 2023 and 2025 files, extracted every relevant formula from your formula sheet.md , and ensured every slide-specific trap is covered.

THE FINAL EXAM TRAP LIST (Review Before Exam)

- **The “Advection vs. Dispersion” Trap:** Do not confuse these! **Molecular Diffusion** is driven by thermal energy (Fick’s Law). **Eddy Dispersion** is driven by turbulent velocity fluctuations. **Advection** is simply the bulk movement of water, calculated as velocity $\times$ concentration ($v \cdot C$).
- **The Natural vs. Enhanced GHG Trap:** Earth would be -18°C without an atmosphere. The trap: **Water Vapour (H_2O)** is the most abundant and potent *natural* greenhouse gas (causing $\sim 50\%$ of the warming). The “enhanced” greenhouse effect is what humans cause via CO_2 and CH_4 .
- **The CMBR Assumption Trap:** If asked for the key assumption of a Batch Reactor (CMBR), the answer is always **“Complete mixing with no inflow or outflow.”**

EVERY Week 4 Concept (Explained Like You’re 5)

Part 1: Mass Balance & Control Volumes

- **1. The Piggy Bank Rule (Mass Balance):** Imagine a piggy bank. The money inside (Accumulation) equals the allowance you put in (Inflow) minus the money you spend on candy (Outflow) plus the interest the bank gives you (Generation/Reaction).
 - *The Trap:* At **Steady State**, nothing is changing over time. Your piggy bank stays exactly the same ($dm/dt = 0$), meaning Input = Output.
- **2. Control Volume:** This is just drawing an imaginary box around the thing you are measuring. It can be any shape or size (a lake, a pipe, a room), as long as it has boundaries where you can count exactly what goes in and out.

Part 2: The Three Types of Reactors

- **3. Batch Reactor (CMBR):** Think of baking a cake. You put all the ingredients into a bowl, mix them perfectly, and stick them in the oven. While it bakes, **nothing goes in, and nothing comes out.**
- **4. Continuous Stirred Tank Reactor (CSTR):** Think of a crowded swimming pool. Someone is pouring a bucket of blue dye into one end, and water is draining out the other end. Because the kids are splashing so much, the dye is **instantaneously completely mixed** everywhere. The water draining out looks exactly the same as the water everywhere inside the pool.
- **5. Plug Flow Reactor (PFR):** Think of a lazy river ride. You get in your tube and float forward. You do not mix with the people ahead of you or behind you. The concentration changes constantly as you move down the length of the river.
 - *The Ultimate Rule:* A PFR gives **higher removal efficiency** in a **much smaller volume** compared to a CSTR.

THE REACTOR FORMULA VAULT

1. Batch Reactor (CMBR) - No flow in or out:

- **Zero-Order:** $C_e = C_0 - kt$
- **First-Order:** $C_e = C_0 e^{-kt}$
- **Second-Order:** $\frac{1}{C_e} = \frac{1}{C_0} + kt$

2. Continuous Stirred Tank Reactor (CSTR) - Steady State:

- **Zero-Order:** $C_{out} = C_0 - kV / Q$
- **First-Order:** $C_{out} = \frac{C_0}{1 + k\theta}$ (Where $\theta = V / Q$ or retention time)

3. Plug Flow Reactor (PFR) - Steady State:

- **First-Order:** $C_e = C_0 e^{-kV / Q}$

Part 3: How Pollution Moves (Transport Processes)

- **6. Advection:** If you drop a leaf in a fast-moving river, the water pushes the leaf downstream. Advection is simply the pollutant moving along with the **bulk fluid flow**.
- **7. Dispersion & Diffusion:** If you drop a drop of food coloring in a still glass of water, it slowly blurs and spreads out.
 - **Molecular Diffusion:** Driven by tiny, random thermal motions. It follows **Fick's Law**, always smoothing out concentration gradients from high to low.
 - **Eddy Dispersion:** In crazy, splashing (turbulent) water, giant swirls of water mix the pollutant way faster than diffusion. This is the dominant spreading mechanism in highly turbulent rivers.

THE 4 WAYS POLLUTANTS MOVE

1. **Advection:** Bulk movement with the flow of the river/air. (Formula: $v \cdot C$).
2. **Molecular Diffusion:** Spreading of a dissolved pollutant due to thermal energy/concentration gradients (Follows Fick's Law).
3. **Eddy Dispersion:** Spreading due to turbulent, random velocity fluctuations in the water.
4. **Gravitational Settling:** For heavy, suspended sediment particles falling out of suspension (Follows Stokes' Law under creeping flow conditions, $Re < 1$).

Part 4: Fluid Mechanics & Energy (The Slide Secrets)

- **8. Reynolds Number (Re):** A special number that tells you how the water is behaving.
 - Smooth, glassy, sheet-like flow = **Laminar** ($Re < 2000$).
 - Crazy, chaotic, splashing flow = **Turbulent** ($Re > 4000$).
 - *Slide Veto Fact:* A "vortex street" around a cylinder occurs perfectly between $Re = 40$ and 1000 .
 - **9. Stokes' Law:** Dropping a marble in honey. Gravity pulls it down (F_g), but buoyancy (the honey holding it up, F_B) and viscous drag (friction, F_D) fight back upward.
 - *Slide Veto Fact:* Stokes' Law only works in "**creeping flow**" conditions, which means the Reynolds number must be **less than 1**.
 - **10. Energy Balance:** Just like mass, energy cannot be destroyed. For a power plant, if you input 2800 MW of fuel and get 1000 MW of electricity, the rest is **Waste Heat**. You calculate waste heat by subtracting the output from the total input.
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The 100% COMPLETE Week 4 Q&A Bank

** Physics**

- **Q: What is the Ideal Gas Law used for in environmental engineering?** [Slide Veto / Formula Sheet]
 - **A:** To calculate gas volumes and pressures. Formula: $P V = nRT$. (Important for air pollution and biogas calculations).
- **Q: Stokes' Law assumes the particles are:** [v2 Guide Spin-off]
 - **A:** Rigid, spherical, and falling in an incompressible fluid at a low concentration (no particle-particle interaction).

Mass Balance & Control Volumes

- **Q:** What is mass flux?

- **A:** The rate at which mass enters and leaves the system.
- **Q:** What is a control volume? **A:** Volume of any shape and location with defined boundaries used for mass flux calculation.
- **Q: What is the primary difference between a “Control Volume” and a “Control Mass”?** [Slide Veto]
 - **A:** A Control Volume is a fixed region in space (water flows through it), while a Control Mass is a fixed amount of matter (like a sealed container).
- **Q:** What is the formula for calculating mass accumulation rate? **A:**

$$\frac{dm}{dt} = \dot{m}_{in} - \dot{m}_{out} + \dot{m}_{reaction}$$
- **Q:** Calculate the mass accumulation rate if 50 mg/L/s is entering, and 10 mg/L/s is leaving (assuming no reaction). **A:** 40 mg/L/s ($50 - 10 = 40$).
- **Q:** Under what condition does the mass accumulation rate (dm/dt) equal zero in a reactor? **A:** Steady-state conditions.

Reactor Theory (CMBR, CSTR, PFR)

- **Q: Match the Reaction Order to its Graphical representation:** [Slide Veto]
 - **Zero-Order:** A plot of **Concentration (C) vs. Time (t)** is a straight line.
 - **First-Order:** A plot of **$\ln(c)$ vs. Time (t)** is a straight line.
- **Q:** What is the key assumption in the mass balance equation for a batch reactor? **A:** Complete mixing with no inflow or outflow.
- **Q:** What is the purpose of the stir bar in a completely mixed batch reactor? **A:** To maintain homogeneous mixing.
- **Q:** In a zero-order batch reactor, how does the concentration change with time? **A:** Linearly (it decreases at a constant rate, independent of concentration).
- **Q:** In which type of reactor are fluid particles instantaneously dispersed throughout the reactor volume? **A:** Continuous Stirred Tank Reactor (CSTR).
- **Q:** How does the concentration change in a CSTR under steady-state conditions? **A:** It remains constant.
- **Q:** Which reactor gives the highest removal efficiency for a given reactor volume? **A:** Plug flow reactor (PFR).
- **Q:** Which of the following is NOT true regarding a plug flow reactor? **A:** Inter mixing of particles will happen. (False: In an ideal PFR, there is NO longitudinal intermixing).
- **Q:** What is the formula for Hydraulic Retention Time (HRT)? **A:** $\theta = V / Q$.

- **Q:** What is the key assumption in the mass balance equation for a batch reactor?
 - **A:** Complete mixing with no inflow or outflow.
- **Q:** Under which condition does the mass accumulation rate (dm/dt) equal zero in a reactor? **A:** Steady-state conditions.
- **Q:** For a CSTR at steady state with a first-order reaction, what is the equation? **A:** $C_{out} = C_0 / (1 + kV / Q)$.
- **Q:** In a batch reactor with a second-order reaction, what describes the effluent concentration? **A:** $1/C_e = 1/C_0 + kt$.
- **Q:** How does concentration change with time in a zero-order batch reactor? **A:** Linearly (decreases at a constant rate).
- **Q:** How does concentration change with time in a first-order batch reactor? **A:** Exponentially.
- **Q:** In the CSTR mass balance, what does the term $\dot{m}_{in}$ equal? **A:** QC_0 .
- **Q:** In a first-order batch reactor, the half-life ($t_{1/2}$) is given by what formula? **A:** $t_{1/2} = \ln(2)/k$.
- **Q:** If the Hydraulic Retention Time (HRT) is doubled while keeping reactor volume constant, what happens to the flow rate? **A:** It halves.
- **Q:** The PFR model assumes that fluid elements behave how? **A:** They move through the reactor in discrete "plugs" with no mixing between them.
- **Q:** Integrating the PFR mass balance for a first-order reaction gives what formula? **A:** $C_e = C_0 e^{-k\theta}$.

Reactor Calculations (Use Your Formula Sheet!)

- **Q:** How long would it take to reduce concentration to e^{-1} ($\approx 36.8\%$) of its initial value in a first-order batch reactor? **A:** $t = 1/k$.
- **Q:** Calculate the time to reduce pollutant A to 1% of its initial concentration in a CMBR (First Order, $k = 0.23 \text{ min}^{-1}$).
 - **A:** 20 min.
 - $\left| \begin{array}{l} \text{Note: } \ln(C_e/C_0) = -kt. \\ \ln(0.01) = -0.23t \rightarrow -4.605 = -0.23t \rightarrow t \approx 20 \text{ min.} \end{array} \right.$

- **Q:** Calculate the flow rate in a CMBR if volume is 100 m^3 and retention time is 5 days.
 - **A:** $20 \text{ m}^3/\text{day}$.
 - | *Note:* $Q = V / \theta = 100/5 = 20 \text{ m}^3/\text{day}$.
- **Q:** Determine the volume of a Plug Flow Reactor (PFR) to obtain an effluent of 50 mg/L . Given: $Q = 100 \text{ m}^3/\text{day}$, $k = 0.23 \text{ day}^{-1}$, $C_{\text{in}} = 200 \text{ mg/L}$.
 - **A:** 603 m^3 .
 - | *Note:* $V = -\frac{Q}{k} \ln\left(\frac{C_{\text{out}}}{C_{\text{in}}}\right) = -\frac{100}{0.23} \ln(0.25) \approx 603 \text{ m}^3$.
- **Q:** For a zero-order CSTR at steady state, what is the outlet concentration?
 - **A:** $C_{\text{out}} = C_0 - kV / Q$.

Transport Processes & Fluid Mechanics

- **Q: Which mass transport process is driven by thermal (Brownian) motion?** [2025]
 - **A:** Molecular Diffusion.
- **Q: True or False: Fick's Law of diffusion assumes the flow velocity is always constant.** [2025]
 - **A: False.** Fick's Law describes diffusion, which occurs due to concentration gradients and is independent of the bulk flow velocity.
- **Q: Why is kinetic energy typically ignored in the energy balance of a municipal water tank?** [2025]
 - **A:** Because the fluid is stationary or moving at a very low, negligible velocity.
- **Q:** Different mass transport processes are?
 - **A:** All the above (Diffusion, Dispersion, and Advection).
- **Q:** What is the fundamental difference between advection and dispersion? **A:** Advection transports pollutants with bulk flow, while dispersion spreads them due to random motion.
- **Q:** Which law is used to calculate dispersive flux density? **A:** Fick's Law.
- **Q:** Which assumption is NOT made in applying Fick's Law to diffusion processes? **A:** Flow velocity is constant (Fick's law describes diffusion, which is independent of bulk advective flow).
- **Q:** What causes turbulent diffusion (eddy dispersion)? **A:** Random fluctuations in the advective velocity.
- **Q:** Which type of pollutant transport is most likely to dominate in highly turbulent water bodies? **A:** Eddy dispersion.

- **Q:** What determines the terminal settling velocity of a particle in a fluid according to Stokes' Law? **A:** Viscosity of the fluid, particle diameter, and density difference between particle and fluid (All of the above).
- **Q:** What is the significance of the drag force in Stokes' law for gravitational settling? **A:** It resists the gravitational force acting on a particle (it acts upward).
- **Q:** Stokes' drag force applies mostly under what specific flow condition? **A:** Creeping flow conditions (Reynolds number less than 1).
- **Q:** Which dimensionless quantity is used to predict if a flow pattern is laminar or turbulent? **A:** Reynolds number.
- **Q:** What is the primary difference between molecular diffusion and eddy dispersion?
 - **A:** Molecular diffusion is caused by thermal energy; eddy dispersion is caused by turbulent velocity fluctuations.
- **Q:** The advective component of the 1D transport equation is represented by:
 - **A:** $\mathbf{v} \cdot \mathbf{C}$ (velocity $\times$ concentration).
- **Q:** Which transport mechanism causes heavy suspended sediment particles to settle out of suspension?
 - **A:** Gravitational settling.
- **Q:** At terminal settling velocity, what is the net downward force on a particle? **A:** Zero (gravity = buoyancy + drag).
- **Q:** The Stokes drag force on a slowly settling sphere is given by what formula? **A:** $F_D = 3\pi\mu D_p v_r$.
- **Q:** What is the formula for buoyancy force (F_B) on a settling particle? **A:** $\rho_f(\pi/6)D_p^3 g$.
- **Q:** If the particle density increases (while fluid stays constant), what happens to the terminal settling velocity? **A:** It increases.
- **Q:** In Stokes' settling, if the particle diameter is doubled, what happens to the settling velocity? **A:** It quadruples ($v_s \propto D_p^2$).
- **Q:** What increases the Reynolds number (making flow more turbulent)? **A:** Increasing the flow velocity (V).
- **Q:** Stokes' gravitational settling assumes what flow condition? **A:** Creeping flow ($Re < 1$).
- **Q:** Which equation relates the energy of a photon to its frequency? **A:** Planck's Equation.
- **Q:** In the energy balance equation for a system at steady state, what is the relationship between Energy in and out? **A:** $E_{in} = E_{out}$ (since $dE/dt = 0$).
- **Q:** What is the formula for heat energy change? **A:** $\Delta E = \text{mass} \times c \times \Delta T$.
- **Q:** Which force on a settling particle acts in the same direction as drag? **A:** Buoyancy force.

- **Q:** Which of the following IS a valid assumption in applying Fick's Law? **A:** Molecular motion is random.
- **Q:** Fick's Law applies even under what specific condition? **A:** When there is zero bulk flow velocity.

Energy Balance & Global Issues

- **Q:** How is energy loss as waste heat typically calculated in a thermal power plant?
A: By subtracting the generator's output energy from the total input energy.
- **Q:** Calculate the change in gravitational energy of a 50 kg object if its height changes from 10 m to 2.5 m ($g = 9.81 \text{ m/s}^2$).
 - **A:** 3679 J.
 - **Note:**

$$\Delta P E = m \times g \times \Delta h = 50 \times 9.81 \times (10 - 2.5) = 3678.75 \approx 3679 \text{ J.}$$
- **Q:** Without an atmosphere and greenhouse gases, what would happen to Earth's surface temperature? **A:** It would be much cooler than today (approx. -18°C).
- **Q:** Which gas acts similarly to a blanket at night, trapping heat? **A:** Greenhouse gases, with water vapor (H_2O) being a primary natural greenhouse gas.
- **Q:** Is it true that without greenhouse gases, Earth's average temperature would be approximately -18°C ? What is the most potent natural GHG?
 - **A:** Yes, true. Water vapour (H_2O) is the most potent natural greenhouse gas.
- **Q:** Which formula is used to calculate Heat internal energy?
 - **A:** Mass $\times$ heat capacity (c) $\times \Delta T$.
- **Q:** Which energy term is irrelevant in a stationary water treatment tank?
 - **A:** Kinetic energy.
 - **Note:** Kinetic energy is associated with bulk movement; since the tank is stationary, it is mathematically zero.

ACTIVE RECALL & FEYNMAN CHALLENGE

Cover the answers below and test yourself before the exam.

Quiz 1: You are using a CSTR (swimming pool) to treat wastewater with a first-order decay reaction. If the initial concentration (C_0) is 100 mg/L, the rate constant (k) is 0.5 day^{-1} , and the retention time (θ) is 2 days, what is the output concentration?

Answer: 50 mg/L. (Formula:

$$C_{\text{out}} = C_0 / (1 + k\theta) \rightarrow 100 / (1 + 0.5 \times 2) \rightarrow 100 / 2 = 50).$$

Quiz 2: If a factory drops a bucket of dissolved red dye into a perfectly still lake with zero current, what transport mechanism will eventually cause the dye to spread out?

***Answer:** Molecular Diffusion (driven by thermal energy and concentration gradients, since there is no turbulent eddy dispersion or advective flow).*

Feynman Challenge: Explain the difference between the “Natural” Greenhouse Effect and the “Enhanced” Greenhouse Effect.

***Example Answer:** The Natural Greenhouse Effect is Earth’s cozy blanket, mostly made of water vapor naturally evaporating from the oceans, keeping us at a livable $+15^{\circ}\text{C}$ instead of a freezing -18°C . The Enhanced Greenhouse Effect is like throwing a second, heavy wool blanket over the bed—caused by humans digging up ancient carbon (fossil fuels) and pumping extra CO_2 and Methane into the air, causing the bed to overheat.*